

Exploratory Data Analysis Report

Marketing Campaign Dataset

Data Analyst Internship – Task 5

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Tools Used: Python, Pandas, Matplotlib, Seaborn, Jupyter Notebook

1. Project Overview

Objective

The objective of this project was to perform Exploratory Data Analysis (EDA) on a marketing campaign dataset to identify customer characteristics, spending behavior, purchasing patterns, and relationships among variables. The analysis aims to transform raw data into meaningful business insights that support informed decision-making.

2. Dataset Information

Item	Description
Dataset	Marketing Campaign
Source	Kaggle
Records	2,240
Features	29
Domain	Marketing & Customer Analytics

The dataset contains customer demographic information, annual income, purchasing behavior across different product categories, and campaign response history.

3. Data Preparation

Before conducting exploratory analysis, the dataset was cleaned to improve data quality and reliability.

The following preprocessing steps were completed:

- Removed duplicate records.
- Handled missing values in the Income column.
- Standardized categorical values.
- Renamed column headers using lowercase with underscores.
- Converted the customer enrollment date into datetime format.
- Corrected inconsistent data types.
- Removed one extreme income outlier (666,666), while retaining valid high-income observations.
- Verified the dataset for completeness and consistency.

The cleaned dataset was then used for all subsequent analyses.

4. Exploratory Data Analysis

Several statistical summaries and visualizations were created to better understand the dataset.

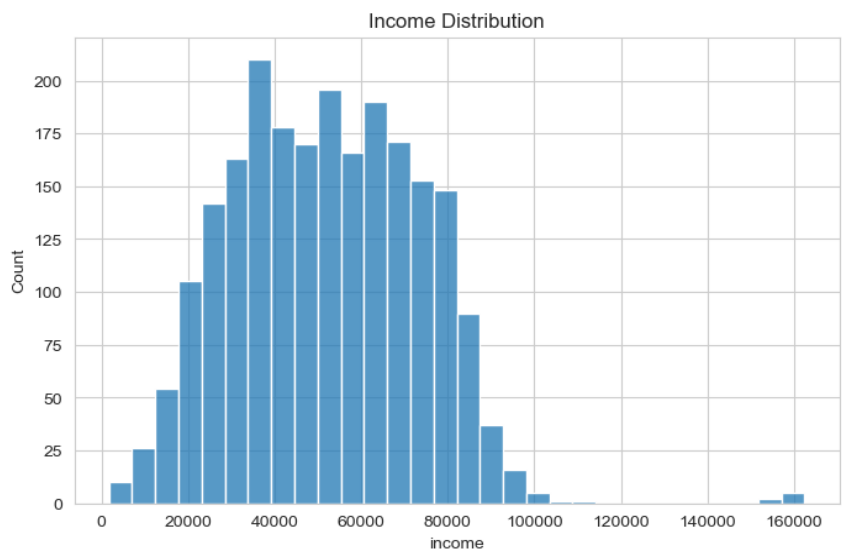
The analyses included:

- Descriptive statistics
 - Distribution analysis
 - Boxplots
 - Histograms
 - Scatter plots
 - Correlation heatmap
 - Pairplot
 - Customer demographic analysis
 - Product spending analysis
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5. Visual Analysis and Findings

Income Distribution

Visualization: Histogram

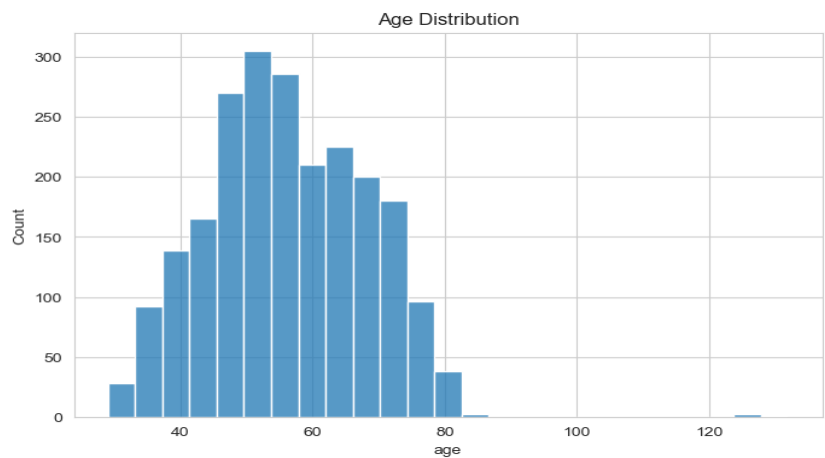


Observation

Most customers earn between 30,000 and 80,000 annually, which represents the main concentration of the distribution. The income pattern shows a slight right-skew, with fewer customers earning above 80,000. Only a very small number of customers fall into the high-income range (above 100,000), with rare outliers reaching around 150,000–160,000.

Age Distribution

Visualization: Histogram

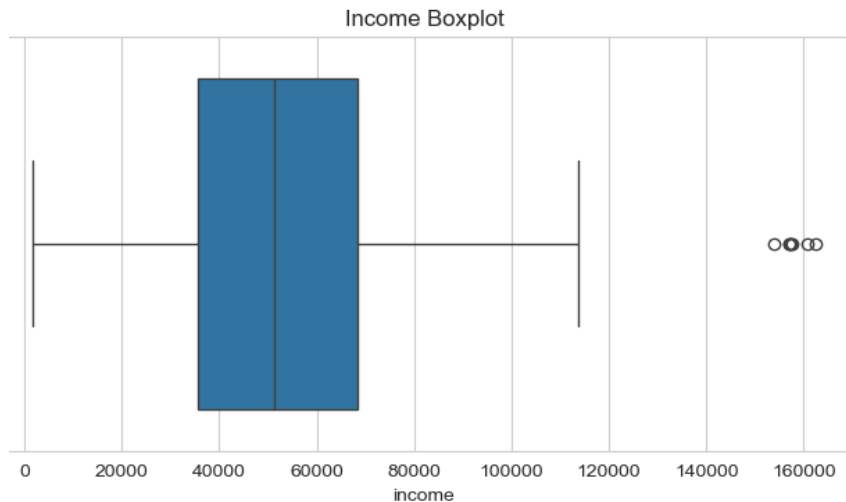


Observation

The majority of customers are middle-aged adults, indicating that the company's customer base primarily consists of economically active individuals.

Income Outliers

Visualization: Boxplot



Observation

Most income values fall within a reasonable range. One unrealistic income value (666,666) was identified and removed during preprocessing, while the remaining high-income observations were retained because they appeared to be valid.

Income vs Wine Spending

Visualization: Scatter Plot

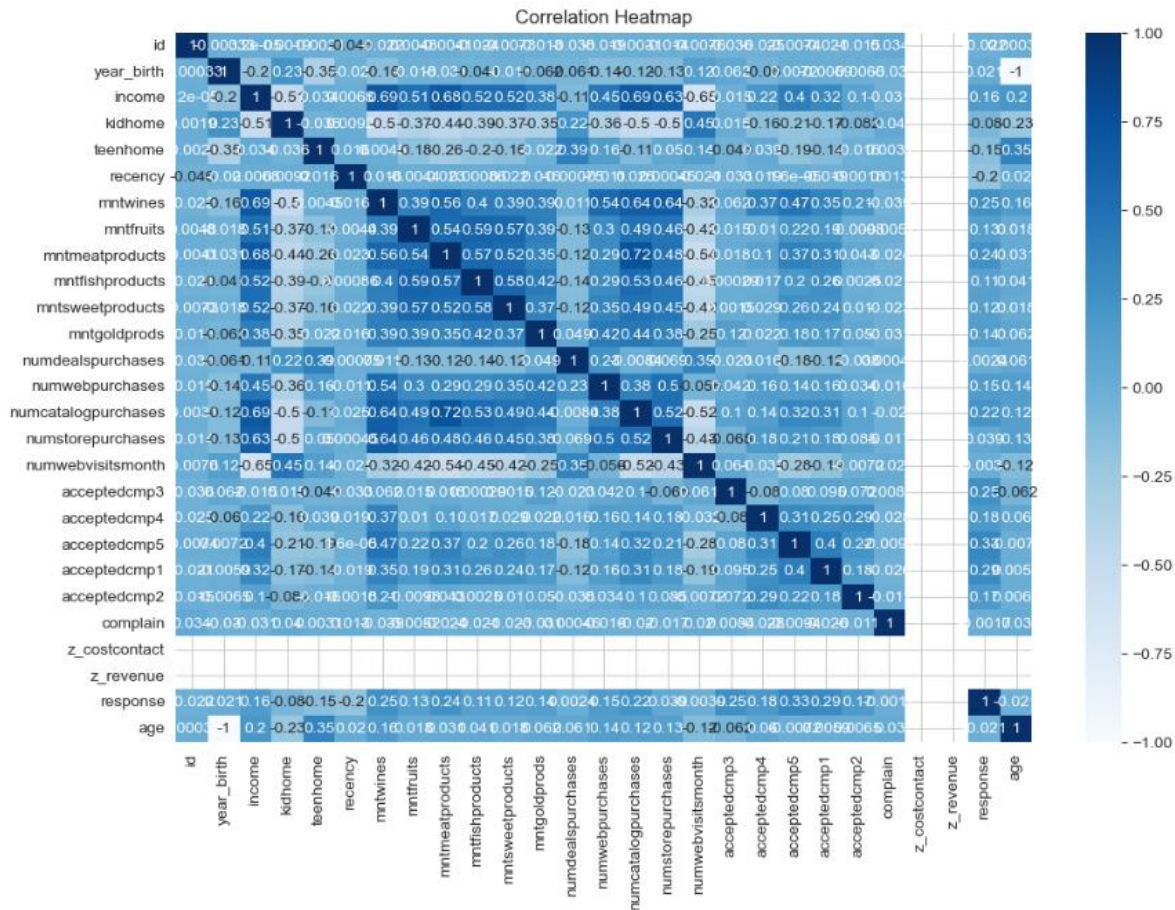


Observation

Customers with higher incomes generally spend more on wine products, suggesting a positive relationship between income and wine purchasing.

Correlation Analysis

Visualization: Correlation Heatmap

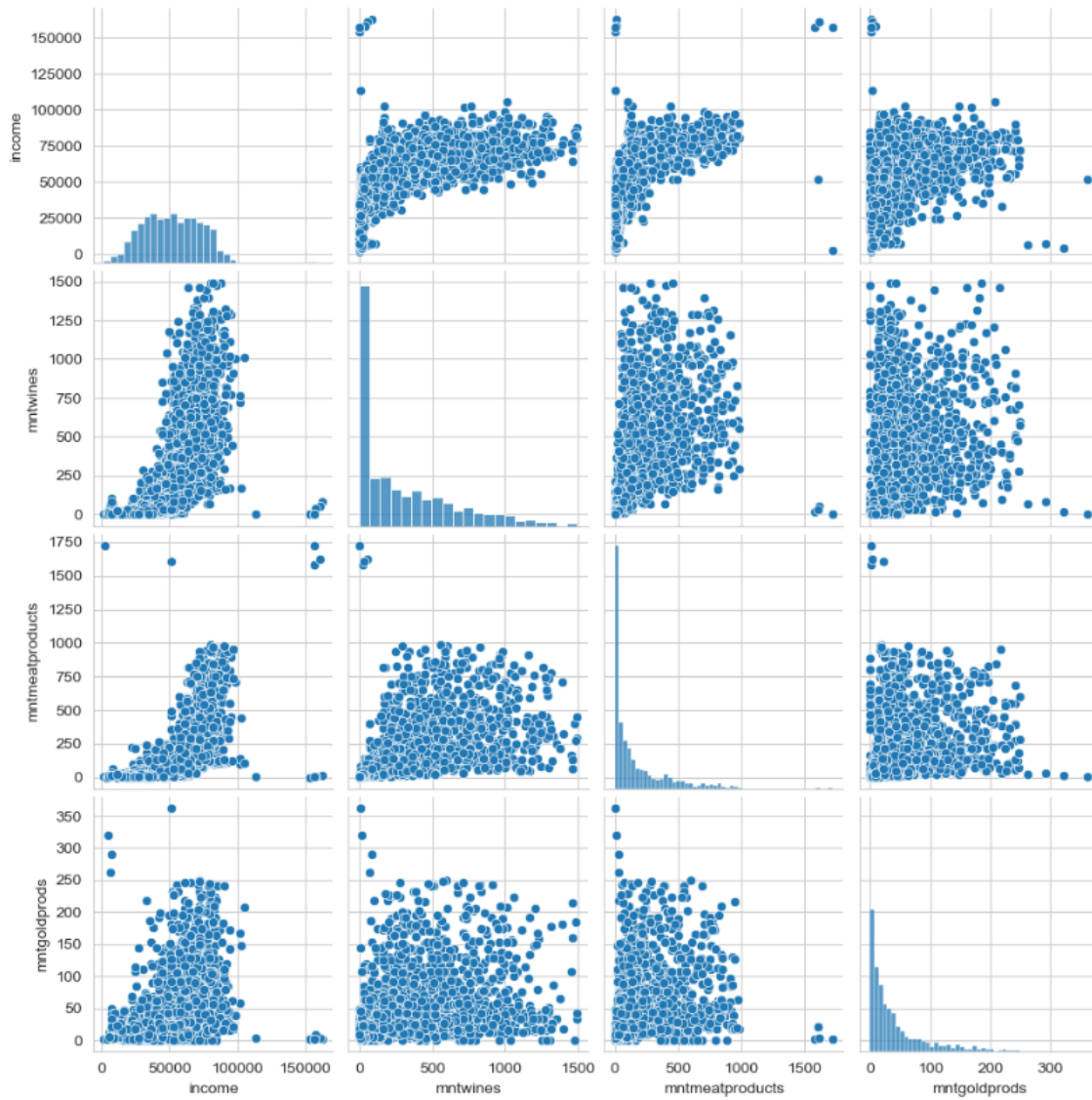


Observation

- Income positively correlates with product spending.
- Spending categories are highly correlated.
- Recency has weak relationships with most variables.

Pairwise Relationships

Visualization: Pairplot

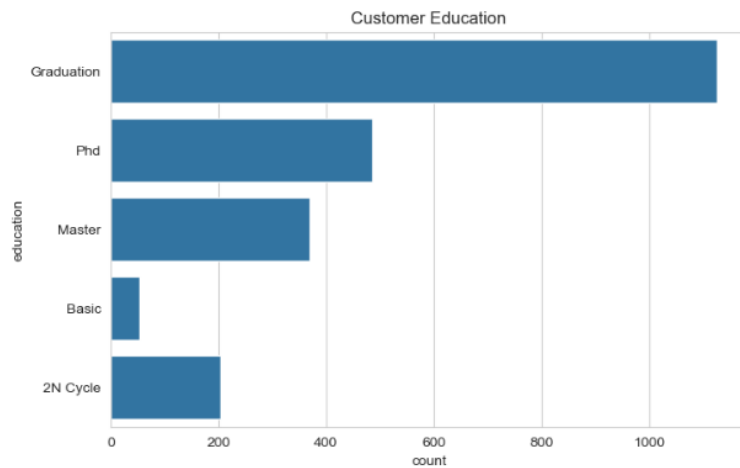


Observation

Customers who spend more on one product category also tend to spend more across other product categories.

Education Distribution

Visualization: Count Plot

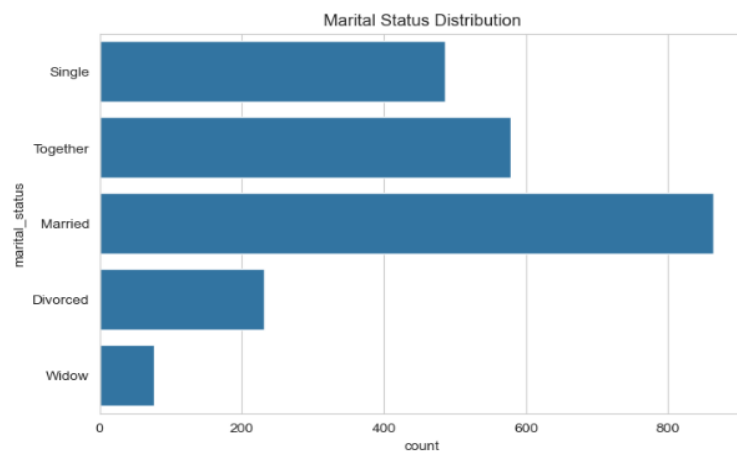


Observation

Graduation is the most common education level among customers, followed by PhD and Master's degree holders.

Marital Status Distribution

Visualization: Count Plot

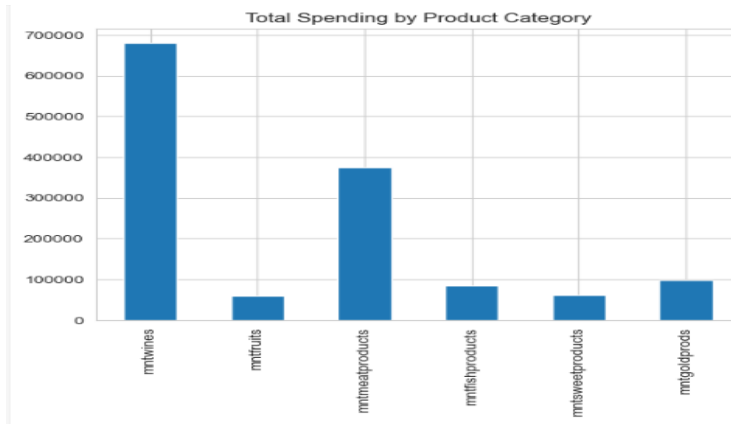


Observation

Married customers represent the largest customer segment, followed by customers living together and single customers.

Product Spending Analysis

Visualization: Bar Chart



Observation

Wine products generated the highest customer spending, while sweet products recorded the lowest overall expenditure.

6. Statistical Summary

Key descriptive statistics were generated using the Pandas `.describe()` function.

The analysis showed:

- Income varies considerably among customers.
- Product spending is highly skewed toward wine purchases.
- Customer recency ranges from recent buyers to customers who have not purchased for several months.
- Spending behavior differs significantly across customers.

7. Business Insights

The exploratory analysis produced several valuable business insights:

- Higher-income customers generally spend more across product categories.
 - Wine represents the company's strongest revenue-generating product category.
 - Graduation is the dominant education level among customers.
 - Married customers form the largest customer group.
 - Product spending behaviors are closely related, indicating opportunities for cross-selling.
 - Customer recency has relatively weak relationships with purchasing variables, suggesting additional factors influence buying behavior.
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8. Conclusion

This Exploratory Data Analysis successfully identified important customer characteristics, spending trends, and relationships within the marketing dataset.

The insights obtained from this analysis can support:

- Customer segmentation
- Targeted marketing campaigns
- Product recommendation strategies
- Revenue optimization
- Customer retention initiatives
- Business decision-making

The cleaned dataset and exploratory findings also provide a strong foundation for future predictive modeling and advanced analytics.

9. Skills Demonstrated

- Data Cleaning
- Exploratory Data Analysis (EDA)
- Data Visualization
- Statistical Analysis
- Correlation Analysis
- Feature Exploration
- Business Insight Generation
- Python Programming
- Pandas
- Matplotlib
- Seaborn
- Jupyter Notebook
- Data Storytelling